

Which Open Model Should the Lab Run?

A consolidated evaluation of 18 open and closed models against the Philips lab constraints, agentic workloads, three orchestration frameworks and the FastAPI AI scaffolding standard.

18

models assessed

6

qualify on scaffolding

1

fits the lab GPU

EVALUATED AGAINST

- Philips MOM — zero external APIs
- A100 80 GB / 2 × L40S envelope
- vLLM self-hosting + LiteLLM gateway
- Single / multi-step / multi-agent
- LangGraph · CrewAI · Agents SDK
- FastAPI ten-point scaffolding harness
- Licence class for enterprise review

Four Gates — How 18 Models Became 1 Recommendation

Each gate is a hard filter derived from the Philips MOM, the lab hardware, or the engineering standard the code must meet.

1

COMPLIANCE GATE

Open weights, self-hostable in a private VPC. Any hosted-only model fails the MOM's zero-external-API rule outright.

Removes: Grok 4.5, Qwen3.7 Max, MiMo-V2.5-Pro

2

HARDWARE GATE

Must serve under vLLM inside 80 GB (A100) or 48 GB per node (L40S), with KV headroom for 21 concurrent seats.

Removes: every model above ~30B active

3

AGENTIC GATE

Reliable structured tool calling across multi-step loops — the actual workload of CrewAI, LangGraph and the Agents SDK.

Removes: StarCoder2, Mistral 7B, all sub-10B

4

SCAFFOLDING GATE

Can hold ten simultaneous engineering constraints and audit its own output honestly against them.

Leaves: a single lab-viable qualifier

The gates are ordered deliberately — compliance and hardware are non-negotiable and cheap to check, so they run before any benchmark comparison. Capability only decides between models that already passed the first two.

What the Philips Lab Actually Demands of a Model

Taken directly from the Lab Infrastructure Proposal v2.0 — the model choice is downstream of all six.

● Zero external LLM APIs

No OpenAI, Anthropic, Google or hosted endpoint. All inference inside the private VPC — no data egress. Weights must be downloadable.

● Fits the GPU envelope

Option A: 1 × A100 80 GB shared by two models (util 0.50 + 0.35). Option B: 2 × L40S 48 GB, one model per node at util 0.90.

● 21 fully parallel seats

vLLM continuous batching with `--max-num-seqs 64` and 4–8K context. KV cache headroom is a real constraint, not a rounding error.

● Served via vLLM + LiteLLM

Model must have mature vLLM support and a clean OpenAI-compatible surface, including tool-call and reasoning parsers.

● Agentic curriculum

CrewAI, LangGraph, LangChain, OpenAI Agents SDK, a participant-built FastAPI MCP server, and OpenCode CLI for spec-driven development.

● Licence fit for enterprise

Apache 2.0 / MIT clears review without friction. Community and OpenRAIL licences carry clauses a Philips-class legal review will read.

Net effect: the compliance and hardware gates together cap the candidate at roughly 30B total parameters — before a single benchmark is consulted.

Frontier Open-Weight Models — Independent Scores

All figures from *llm-stats.com* (composite blends verified benchmarks, live API metrics and pricing; revalidated hourly).

Model	Params (B)	LLM Stats	Coding	Tools	SWE-Pro	SWE-V	Context	Latency	\$/M in-out	Licence	Fits lab GPU?
Kimi K3	2,800	55.6	44.6	39.2	—	—	1.0M	14.2s	3.00 / 15.00	Weights Jul 27	No
GLM-5.2	753	47.6	39.8	26.1	62.1%	—	1.0M	5.1s	0.95 / 3.00	MIT	No
Kimi K2.6	1,000	44.8	35.4	23.5	58.6%	80.2%	262K	18.0s	0.75 / 3.50	Mod. MIT	No
DeepSeek V4 Pro	1,600	44.1	33.8	25.4	55.4%	80.6%	1.0M	11.0s	1.60 / 3.20	Open	No
Hy3 (Tencent)	295	43.9	35.8	25.6	57.9%	78.0%	—	—	—	Hunyuan Comm.	No
GLM-5.1	754	40.8	33.5	21.6	58.4%	—	200K	2.0s	1.40 / 4.40	MIT	No
DeepSeek V4 Flash	284	39.8	29.4	19.5	52.6%	79.0%	1.0M	4.6s	0.10 / 0.20	Open	No
Qwen3.6-27B	27.8	36.7	27.1	15.3	53.5%	77.2%	262K	4.7s	0.60 / 3.60	Apache 2.0	YES
MiniMax M2.1	230	33.0	22.8	16.2	—	67.0%	1.0M	1.8s	0.30 / 1.20	Community	No

Best open model overall: GLM-5.2 — top open-weights composite, highest open SWE-bench Pro (62.1%), 1M context and the lowest frontier price.

Best model per parameter: Qwen3.6-27B — beats DeepSeek V4 Flash on SWE-bench Pro (53.5 vs 52.6) at one-tenth the parameter count.

The Rest of the Field — Closed, Small, and Disqualified

Two separate reasons for exclusion: hosted-only models fail compliance; sub-10B models fail the agentic workload.

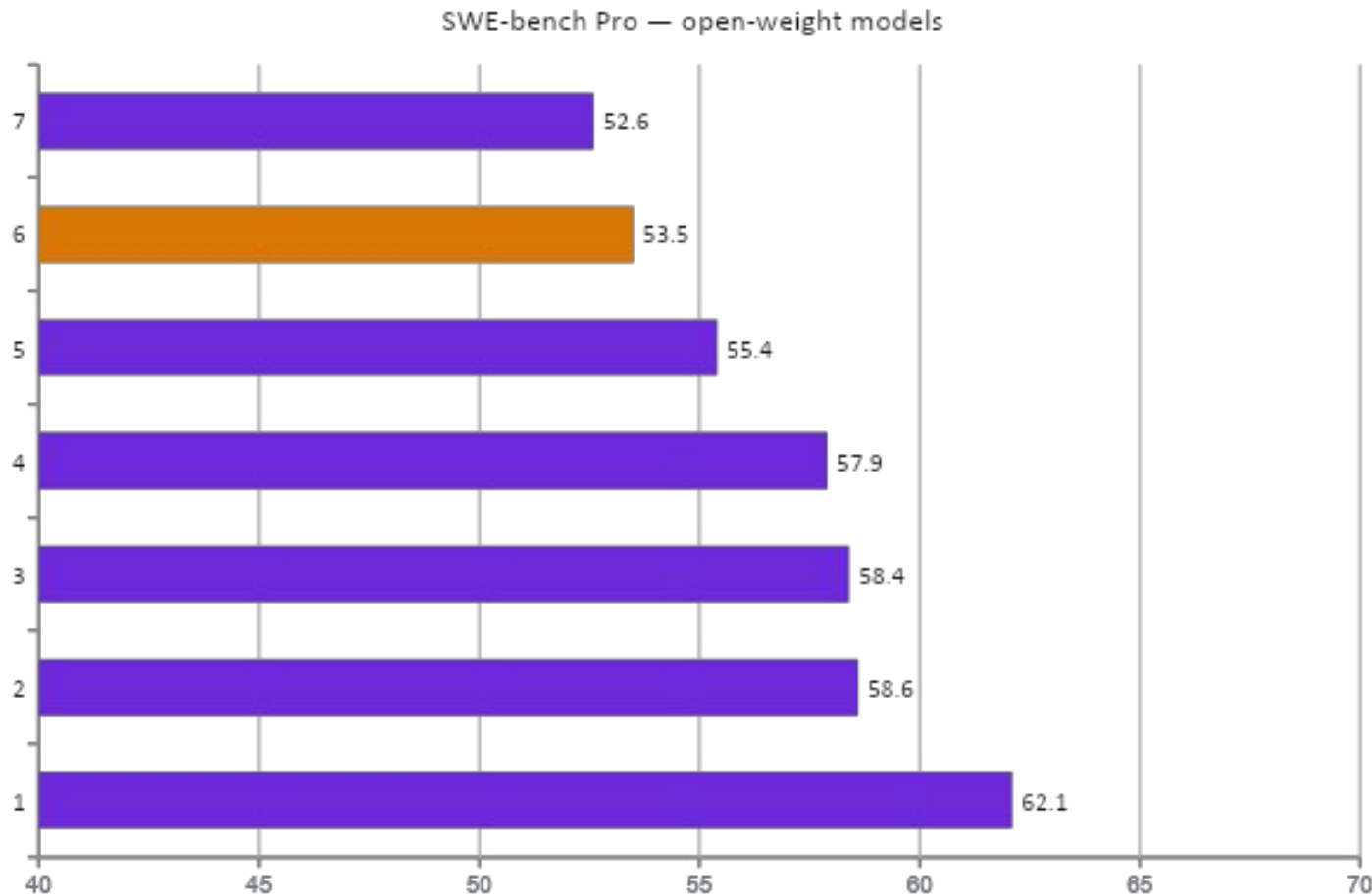
Model	Class	LLM Stats	Coding	Licence	Why it is out of scope for Philips
Grok 4.5 (xAI)	Closed frontier	49.0	39.9	Proprietary	Hosted API only — violates the zero-external-API clause of the MOM.
Qwen3.7 Max	Closed frontier	46.8	39.1	Proprietary	Hosted only. Strong agentic endurance, but no self-host path exists.
MiMo-V2.5-Pro	Closed	—	—	Proprietary	Listed as proprietary and served via hosted providers only.
Qwen3.5-9B	Small open	25.7	no index	Apache 2.0	No coding index recorded — the sub-10B tier does not chart on agentic work.
Gemma 4 26B-A4B	Small open	30.4	—	Gemma ToS	Viable size, but no coding or SWE index; unproven for agent loops.
Mistral 7B Instruct v0.3	Small open	not ranked	—	Apache 2.0	CURRENT LAB MODEL — persistent tool-call format errors in multi-turn agent loops; pre-dates mature Pydantic v2 idiom.
StarCoder2-7B	Base / completion	not ranked	—	OpenRAIL-M	CURRENT LAB MODEL — base completion model: no chat template, no tool-calling training, cannot drive an agent at all.

The uncomfortable finding

Both models currently specified in the lab proposal sit below the ranking threshold of the independent leaderboard, and both fail the agentic gate. They were chosen to fit the GPU envelope — which they do, comfortably — but the envelope now admits models several tiers stronger. This is a correctable specification gap, not a design flaw: the topology is model-agnostic by construction.

Capability per Parameter: Why 27B Is the Sweet Spot

SWE-bench Pro measures repo-level engineering discipline — the closest public proxy for the scaffolding workload.



27.8B

parameters for 53.5% SWE-bench Pro

Qwen3.6-27B lands within 8.6 points of the best open model in the world while running on one L40S. GLM-5.2 needs roughly 27× the parameters for that lead.

What fits the lab GPU envelope

- Qwen3.6-27B (4-bit) ~16 GB
- Devstral-24B (AWQ) ~14 GB
- Qwen2.5-Coder-7B ~15 GB
- DeepSeek V4 Flash ~150 GB
- GLM-5.2 8 × H200 class
- Kimi K3 / V4 Pro multi-node cluster

Single-Step, Multi-Step Loop, and Multi-Agent Orchestration

Three tiers reward different properties — which is why no single model sweeps the board.

Model	Single-step generate / edit	Multi-step loop plan → test → fix	Multi-agent orchestration	Best role in a multi-agent system
Kimi K3	★★★★★	★★★★★	★★★★☆	Deep-reasoning specialist node — latency rules out chatty orchestration
GLM-5.2	★★★★★	★★★★☆	★★★★☆	Supervisor / planner — fast enough to drive a whole graph
Kimi K2.6	★★★★☆	★★★★★	★★★★★	Long-session orchestrator (Agent Swarm); 18s latency is the cost
DeepSeek V4 Pro	★★★★★	★★★★☆	★★★☆☆	Escalation node for the hardest reasoning steps
Hy3	★★★★☆	★★★☆☆	★★★☆☆	General worker; strong reasoning per dollar
DeepSeek V4 Flash	★★★★☆	★★★☆☆	★★★★☆	Worker fleet — cost dominates the bill in high fan-out designs
Qwen3.6-27B	★★★★☆	★★★☆☆	★★★☆☆	Long-context worker node — the lab's viable BUILD model
MiniMax M2.1	★★★☆☆	★★★☆☆	★★★★☆	Cheap specialist agents behind a stronger manager
Mistral 7B	★★★☆☆	★★★☆☆	★★★☆☆	Leaf node only — classify, summarise, route. Never a planner.
StarCoder2-7B	★★★☆☆	—	—	Autocomplete engine behind a tool wrapper — no agentic role

Multi-agent orchestration rewards two different things at once — a smart planner and cheap, fast workers. In a high fan-out design, worker cost dominates the bill, not planner intelligence. That is the orchestrator / worker split, and it is exactly the staffing logic of a human team.

LangGraph · CrewAI · OpenAI Agents SDK

Each framework demands a different property from the model — API compatibility is not the differentiator.

LangGraph

Demands STATE DISCIPLINE

Predictable behaviour inside cycles, clean checkpoint / resume, tolerance of long accumulated histories.

Best fit

GLM-5.2 · Kimi K2.7 Code

CrewAI

Demands ROLE ENDURANCE

Staying in character across many delegated tasks; max_iter is the cost driver — set 5–8 per agent, not the default 25.

Best fit

Kimi K2.6 · MiniMax M2.1 as workers

OpenAI Agents SDK

Demands TOOL-CALL RELIABILITY

Handoffs ARE function calls, so a fumbled schema breaks the routing itself. No built-in checkpointing.

Best fit

DeepSeek V4 Flash · GLM-5.2

The failure mode that costs teams two days

Reasoning-tag leakage. Self-hosted Qwen and GLM models emit <think> traces that must be stripped by the correct vLLM reasoning parser and tool-call parser. Misconfigured, the frameworks see the trace as message content and tool calls silently fail — the agent stops mid-task with no error. Hosted APIs handle this for you; a VPC deployment does not.

ACTION FOR THE RUNBOOK

Extend pre-flight checks 2 and 3 to assert a structured tool call round-trips — not merely that a chat completion returns.

The FastAPI Ten-Point Scaffolding Harness

The code participants write must pass all ten checks. That tests sustained constraint-holding, not raw coding ability.

- 1 Type hints on every boundary
- 2 LLM output → Pydantic v2 strict
- 3 Async I/O; loop never blocked
- 4 Typed client, retries + backoff
- 5 Provider behind a port
- 6 Tool args validated pre-execution
- 7 Destructive tools: rule + auth guards
- 8 Request AND response models
- 9 Contract tests as a CI gate
- 10 Core imports no SDK, SQL, framework

What this actually tests in a model

- | | |
|--------------------------------|---|
| Long-context adherence | all ten constraints must survive to the last file generated |
| Current Python idiom | Pydantic v2, not v1 — stale training fails check 2 automatically |
| Honest self-audit | "I assume it's fine" is a FAIL — weak models stamp PASS on everything |
| Agentic file discipline | actually read the reference, grep the core, run the tests |

QUALIFIES

- GLM-5.2 — best available today
- Kimi K3 — batch audits only
- DeepSeek V4 Pro
- Kimi K2.6 · Hy3 · GLM-5.1
- Qwen3.6-27B — lab tier, BUILD mode
- V4 Flash — bounded tasks only

DISQUALIFIED

- StarCoder2-7B — cannot enter BUILD mode at all
- Mistral 7B — drops constraints; Pydantic v1 idiom
- Gemma 2B — local demo role only
- All sub-10B — AUDIT mode collapses into false PASS

Recommended Change to the Lab Model Stack

Same hardware, same cost envelope, same VPC story — cleaner licences and a materially better lab experience.

Slot in the golden image	Currently specified	Recommended	Why it matters for this cohort
DEFAULT_MODEL reasoning / agent driver	Mistral 7B Instruct v0.3 Apache 2.0 · ~15 GB	Qwen3.6-27B (4-bit) Apache 2.0 · ~16 GB	Native tool-call template and current Python idiom. SWE-bench Pro 53.5% vs unranked. Conservative fallback: Qwen3-8B.
CODE_MODEL OpenCode / scaffolding	StarCoder2-7B OpenRAIL-M · ~15 GB	Devstral-24B (AWQ) Apache 2.0 · ~14 GB	Purpose-built for software-agent loops. StarCoder2 is a base completion model with no tool calling — it cannot drive OpenCode.
LOCAL_MODEL Ollama on each seat	Gemma 2B (CPU)	Keep unchanged	Its job is the "run a model on your own machine" moment in S0 and the degraded-mode fallback. Perfect as specified.
Licence OpenRAIL-M use restrictions drop out — both models Apache 2.0. Cleaner enterprise review. Cost Zero change. Same GPU envelope, same \$295–410 two-day total. Blast radius config.yaml + .env only. Runbook, pre-flight script and topology unchanged.			

Optional teaching keeper: run one exercise on Mistral 7B so participants watch it drop constraints 4 onward — then watch the audit catch it. The harness, not the model, is what you trust.

Model Evaluation for the Philips Agentic AI Lab · ArchitectsVibe / Lakshminarayanan for UNext · Sources: llm-stats.com (Jul 2026) · Confidential 9

One Recommendation per Deployment Tier

The lab topology is model-agnostic by construction — the same architecture carries Philips from tier 1 to tier 3 with a config change.

TIER 1

THE BOOTCAMP LAB

Qwen3.6-27B + Devstral-24B

1 × A100 80 GB or 2 × L40S 48 GB

- Fits the existing GPU envelope at 4-bit
- Apache 2.0 — no licence review friction
- 53.5% SWE-bench Pro at 27.8B parameters
- Passes the scaffolding gate in BUILD mode

\$295–410 for two days

TIER 2

PHILIPS PRODUCTION PILOT

DeepSeek V4 Flash

4 × A100 / H100 or 2 × H200

- Smallest genuinely frontier-class open model
- 79.0% SWE-bench Verified at 13B active
- \$0.10 / \$0.20 per M — cheapest serious coder
- Official vLLM support; plugs into OpenCode

One model swap from tier 1

TIER 3

ENTERPRISE SCALE

GLM-5.2

8 × H200-class cluster

- Best open-weights model on the leaderboard
- Highest open SWE-bench Pro at 62.1%
- MIT licence — zero legal review friction
- 1M context; 5.1s latency keeps loops usable

Same gateway topology

Kimi K3 is the capability ceiling and worth re-evaluating once its weights land — but 14.2s latency makes it a batch-audit specialist, not an interactive lab model.

What Stands Out for Philips

One model clears all four gates on the hardware the proposal already specifies.

Qwen3.6-27B

Apache 2.0 · 27.8B parameters · 262K context · 4.7s latency

● Compliance

Fully open weights, self-hosted under vLLM in the private VPC. Zero external calls.

● Hardware

~16 GB at 4-bit — fits one L40S node or shares the A100 with a code model.

● Capability

53.5% SWE-bench Pro and 77.2% SWE-bench Verified — beats DeepSeek V4 Flash at 1/10th the size.

● Scaffolding

Holds the ten checks in BUILD mode when working from the reference implementation.

Paired with Devstral-24B for the code slot

Apache 2.0, ~14 GB at AWQ, purpose-built for agentic software engineering — the model behind OpenCode should be able to plan, call tools and edit files. StarCoder2 cannot. This pairing gives 21 seats a lab where the frameworks behave the way the curriculum says they behave.

AND THE HONEST CAVEAT

Qwen3.6-27B's tools index (15.3) sits well below the frontier qualifiers. It is a strong BUILD model and a capable worker node — it is not an auditor. Final AUDIT verdicts against the ten checks should stay with the trainer, or with a batched GLM-5.2 pass if Philips provisions a larger node.

Decisions and Validation Before T-4

Four items — none of which change the topology, the cost envelope or the provisioning timeline.

1

Confirm the model swap with UNext and Philips

Replace Mistral 7B / StarCoder2-7B with Qwen3.6-27B / Devstral-24B in the golden image. Both Apache 2.0 — a simpler licence story than the current OpenRAIL-M dependency.

Trainer + UNext

T-5

2

Validate quantised variants and KV headroom

Serve both models on a scratch instance; confirm AWQ / 4-bit availability and 21-way concurrency at --max-num-seqs 64. Fallback ladder: Qwen3-8B + Qwen2.5-Coder-7B-Instruct.

UNext

T-4

3

Add a tool-call assertion to pre-flight

Checks 2 and 3 currently assert a chat completion returns. Extend them to assert a structured tool call round-trips, and configure the vLLM tool-call and reasoning parsers explicitly.

UNext + Trainer

T-3

4

Re-evaluate Kimi K3 after the weight release

Weights were announced for 27 July. Out of scope for this delivery on latency and size, but relevant to any Philips production-tier conversation that follows.

Trainer

Post-delivery

Nothing above changes the runbook's commands, the guardrail architecture, the \$295–410 envelope, or the T-24-hour hard gate.